

STAR-FORMING NEBULAE

Star formation can be seen throughout the Milky Way, but it is principally evident in the spiral arms and toward the galactic center, where there is an abundance of star-making ingredients: dust and gas. In these regions, the interstellar matter is dense enough for molecular clouds to exist. These clouds are cold and appear as dark nebulae that are visible only

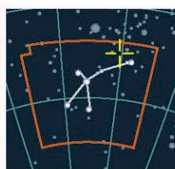
STELLAR NURSERY

Bright young stars within the Omega Nebula, M17, light up the nebula from which they were born.

when framed against a brighter background. When stars are born, these clouds are illuminated from within to become emission nebulae, some of the most beautiful sights in the Milky Way.

DARK NEBULA

BHR 71



CATALOG NUMBER
BHR 71

DISTANCE FROM SUN
600 light-years

MUSCA

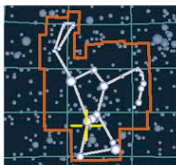
The small dark nebula BHR 71 is called a Bok globule (see p.238) and has a diameter of about 1 light-year. Within the dark molecular cloud are two sources of infrared and radio rays believed to be very close embryonic stars: HH 320 and HH 321, both losing vast amounts of material as they collapse. HH 320 has the strongest outflow, and it is probably surrounded by a massive disk of previously ejected stellar material. Although not optically visible, HH 320 has 10 times the luminosity of the Sun. BHR 71 and its protostars offer a rare opportunity for the study of star-formation processes.

BINARY FORMATION

Jets from BHR 71's newly forming binary star system have created the filamentary structure seen in this composite image made from four separate images.

DARK NEBULA

Horsehead Nebula



CATALOG NUMBER
Barnard 33

DISTANCE FROM SUN
1,500 light-years

ORION

One of the most beautiful and well-known astronomical sights, the Horsehead Nebula can be located in the night sky just south of the bright star Zeta (ζ) Orionis, the left star of the three in Orion's belt (see pp.390–391). The nebula is an extremely dense, cold, dark cloud of gas and dust, silhouetted against the bright, active nebula IC 434. It is about 16 light-years across and has a total mass about 300 times that of the Sun. The Horsehead shape is sculpted out of dense interstellar material by the radiation from the hot young star Sigma (σ) Orionis. Within the dark cloud from which the Horsehead rears is a scattering of young stars in the process of forming. The streaks that extend through the bright area above the Horsehead are probably caused by magnetic fields within the nebula.



DARK KNIGHT

One of the most photographed objects in the night sky, this dark nebula resembles the head of a sea horse or a knight on a chessboard. Its unusual shape was first discovered on a photographic plate in 1888.